

Thermodynamics

Instructor

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Chapter 9 – Problems

**Problem numbers correspond to
5th Edition**

9-14 An air-standard cycle with variable specific heats is executed in a closed system and is composed of the following four processes:

1-2 Isentropic compression from 100 kPa and 27°C to 800 kPa

2-3 $\nu = \text{constant}$ heat addition to 1800 K

3-4 Isentropic expansion to 100 kPa

4-1 $P = \text{constant}$ heat rejection to initial state

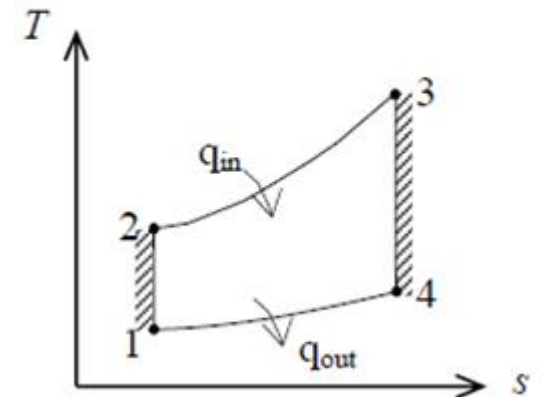
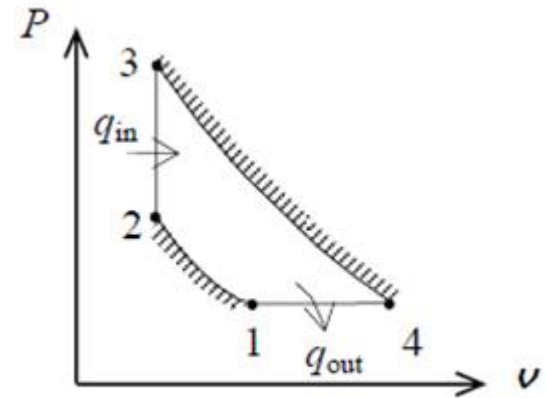
(a) Show the cycle on P - ν and T - s diagrams.

(b) Calculate the net work output per unit mass.

(c) Determine the thermal efficiency.



Solution



Analysis (b) The properties of air at various states are

$$T_1 = 300\text{K} \longrightarrow \begin{aligned} h_1 &= 300.19 \text{ kJ/kg} \\ P_{r_1} &= 1.386 \end{aligned}$$

$$P_{r_2} = \frac{P_2}{P_1} P_{r_1} = \frac{800 \text{ kPa}}{100 \text{ kPa}} (1.386) = 11.088 \longrightarrow \begin{aligned} u_2 &= 389.22 \text{ kJ/kg} \\ T_2 &= 539.8 \text{ K} \end{aligned}$$

$$T_3 = 1800 \text{ K} \longrightarrow \begin{aligned} u_3 &= 1487.2 \text{ kJ/kg} \\ P_{r_3} &= 1310 \end{aligned}$$

$$\frac{P_3 v_3}{T_3} = \frac{P_2 v_2}{T_2} \longrightarrow P_3 = \frac{T_3}{T_2} P_2 = \frac{1800 \text{ K}}{539.8 \text{ K}} (800 \text{ kPa}) = 2668 \text{ kPa}$$

$$P_{r_4} = \frac{P_4}{P_3} P_{r_3} = \frac{100 \text{ kPa}}{2668 \text{ kPa}} (1310) = 49.10 \longrightarrow h_4 = 828.1 \text{ kJ/kg}$$

From energy balances,

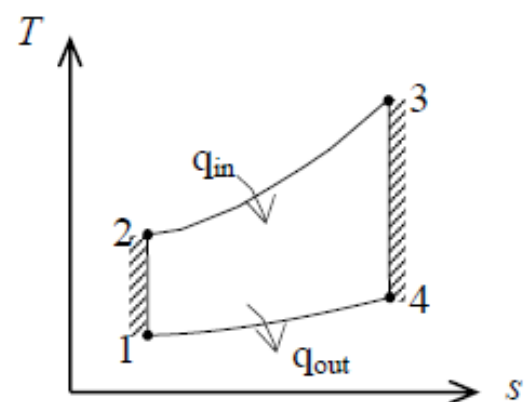
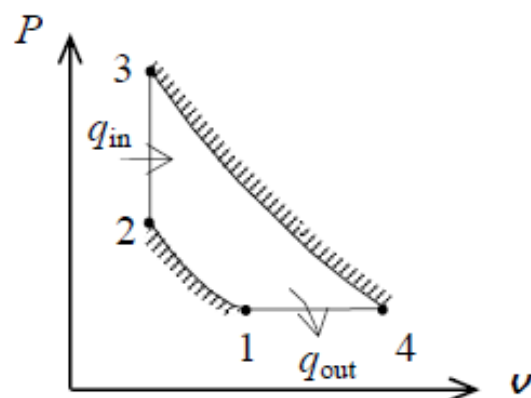
$$q_{\text{in}} = u_3 - u_2 = 1487.2 - 389.2 = 1098.0 \text{ kJ/kg}$$

$$q_{\text{out}} = h_4 - h_1 = 828.1 - 300.19 = 527.9 \text{ kJ/kg}$$

$$w_{\text{net,out}} = q_{\text{in}} - q_{\text{out}} = 1098.0 - 527.9 = \mathbf{570.1 \text{ kJ/kg}}$$

(c) Then the thermal efficiency becomes

$$\eta_{\text{th}} = \frac{w_{\text{net,out}}}{q_{\text{in}}} = \frac{570.1 \text{ kJ/kg}}{1098.0 \text{ kJ/kg}} = \mathbf{51.9\%}$$



9-16 An air-standard cycle is executed in a closed system and is composed of the following four processes:

1-2 Isentropic compression from 100 kPa and 27°C to 1 MPa

2-3 $P = \text{constant}$ heat addition in amount of 2800 kJ/kg

3-4 $v = \text{constant}$ heat rejection to 100 kPa

4-1 $P = \text{constant}$ heat rejection to initial state

(a) Show the cycle on P - v and T - s diagrams.

(b) Calculate the maximum temperature in the cycle.

(c) Determine the thermal efficiency.

9-19 An air-standard cycle is executed in a closed system with 0.004 kg of air and consists of the following three processes:

- 1-2 Isentropic compression from 100 kPa and 27°C to 1 MPa
- 2-3 $P = \text{constant}$ heat addition in the amount of 2.76 kJ
- 3-1 $P = c_1 v + c_2$ heat rejection to initial state (c_1 and c_2 are constants)

- (a) Show the cycle on P - v and T - s diagrams.
- (b) Calculate the heat rejected.
- (c) Determine the thermal efficiency.

Assume constant specific heats at room temperature.

Answers: (b) 1.679 kJ, (c) 39.2 percent

9-20 An air-standard cycle with variable specific heats is executed in a closed system with 0.003 kg of air and consists of the following three processes:

- 1-2 $v = \text{constant}$ heat addition from 95 kPa and 17°C to 380 kPa
- 2-3 Isentropic expansion to 95 kPa
- 3-1 $P = \text{constant}$ heat rejection to initial state

- (a) Show the cycle on P - v and T - s diagrams.
- (b) Calculate the net work per cycle, in kJ.
- (c) Determine the thermal efficiency.

9-21 Repeat Problem 9-20 using constant specific heats at room temperature.

Analysis (b) From the isentropic relations and energy balance,

$$\frac{P_2 v_2}{T_2} = \frac{P_1 v_1}{T_1} \longrightarrow T_2 = \frac{P_2}{P_1} T_1 = \frac{380 \text{ kPa}}{95 \text{ kPa}} (290 \text{ K}) = 1160 \text{ K}$$

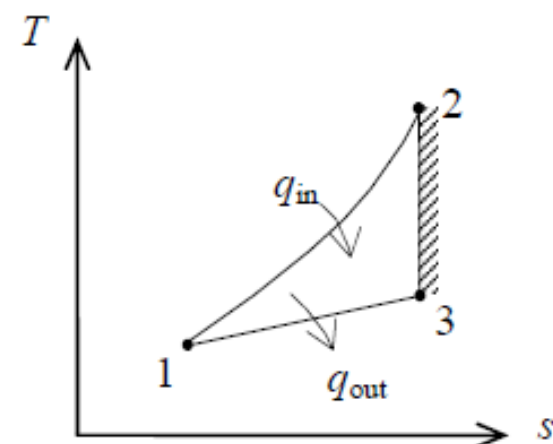
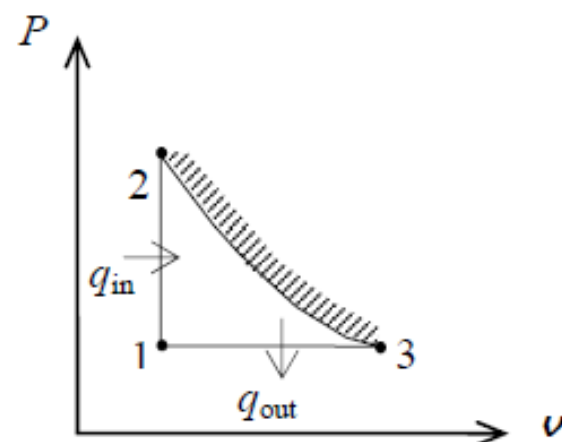
$$T_3 = T_2 \left(\frac{P_3}{P_2} \right)^{(k-1)/k} = (1160 \text{ K}) \left(\frac{95 \text{ kPa}}{380 \text{ kPa}} \right)^{0.4/1.4} = 780.6 \text{ K}$$

$$\begin{aligned} Q_{\text{in}} &= m(u_2 - u_1) = mc_v(T_2 - T_1) \\ &= (0.003 \text{ kg})(0.718 \text{ kJ/kg} \cdot \text{K})(1160 - 290) \text{ K} = 1.87 \text{ kJ} \end{aligned}$$

$$\begin{aligned} Q_{\text{out}} &= m(h_3 - h_1) = mc_p(T_3 - T_1) \\ &= (0.003 \text{ kg})(1.005 \text{ kJ/kg} \cdot \text{K})(780.6 - 290) \text{ K} = 1.48 \text{ kJ} \end{aligned}$$

$$W_{\text{net,out}} = Q_{\text{in}} - Q_{\text{out}} = 1.87 - 1.48 = \mathbf{0.39 \text{ kJ}}$$

$$(c) \quad \eta_{\text{th}} = \frac{W_{\text{net}}}{Q_{\text{in}}} = \frac{0.39 \text{ kJ}}{1.87 \text{ kJ}} = \mathbf{20.9\%}$$

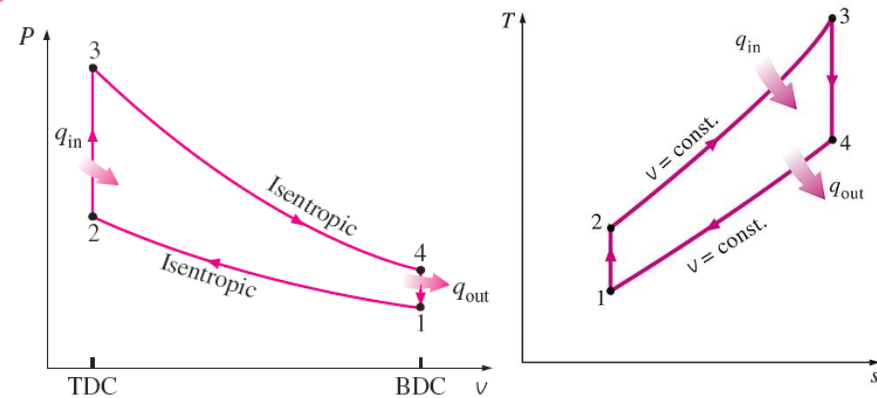


9-22 Consider a Carnot cycle executed in a closed system with 0.003 kg of air. The temperature limits of the cycle are 300 and 900 K, and the minimum and maximum pressures that occur during the cycle are 20 and 2000 kPa. Assuming constant specific heats, determine the net work output per cycle.

9-25 Consider a Carnot cycle executed in a closed system with air as the working fluid. The maximum pressure in the cycle is 800 kPa while the maximum temperature is 750 K. If the entropy increase during the isothermal heat rejection process is $0.25 \text{ kJ/kg} \cdot \text{K}$ and the net work output is 100 kJ/kg, determine (a) the minimum pressure in the cycle, (b) the heat rejection from the cycle, and (c) the thermal efficiency of the cycle. (d) If an actual heat engine cycle operates between the same temperature limits and produces 5200 kW of power for an air flow rate of 90 kg/s, determine the second law efficiency of this cycle.

Otto Cycle

9-37 The compression ratio of an air-standard Otto cycle is 9.5. Prior to the isentropic compression process, the air is at 100 kPa, 35°C, and 600 cm³. The temperature at the end of the isentropic expansion process is 800 K. Using specific heat values at room temperature, determine (a) the highest temperature and pressure in the cycle; (b) the amount of heat transferred in, in kJ; (c) the thermal efficiency; and (d) the mean effective pressure. *Answers: (a) 1969 K, 6072 kPa, (b) 0.59 kJ, (c) 59.4 percent, (d) 652 kPa*



Analysis (a) Process 1-2: isentropic compression.

$$T_2 = T_1 \left(\frac{\nu_1}{\nu_2} \right)^{k-1} = (308 \text{ K})(9.5)^{0.4} = 757.9 \text{ K}$$

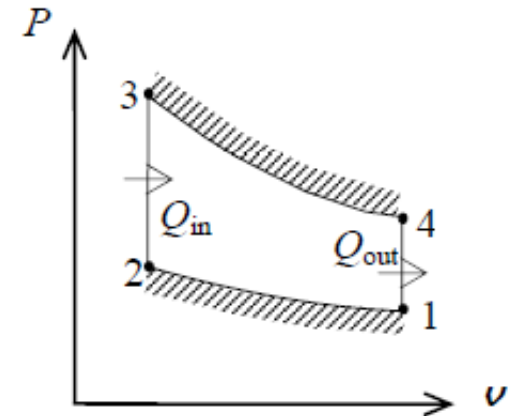
$$\frac{P_2 \nu_2}{T_2} = \frac{P_1 \nu_1}{T_1} \longrightarrow P_2 = \frac{\nu_1}{\nu_2} \frac{T_2}{T_1} P_1 = (9.5) \left(\frac{757.9 \text{ K}}{308 \text{ K}} \right) (100 \text{ kPa}) = 2338 \text{ kPa}$$

Process 3-4: isentropic expansion.

$$T_3 = T_4 \left(\frac{\nu_4}{\nu_3} \right)^{k-1} = (800 \text{ K})(9.5)^{0.4} = \mathbf{1969 \text{ K}}$$

Process 2-3: $\nu = \text{constant}$ heat addition.

$$\frac{P_3 \nu_3}{T_3} = \frac{P_2 \nu_2}{T_2} \longrightarrow P_3 = \frac{T_3}{T_2} P_2 = \left(\frac{1969 \text{ K}}{757.9 \text{ K}} \right) (2338 \text{ kPa}) = \mathbf{6072 \text{ kPa}}$$



$$(b) \quad m = \frac{P_1 V_1}{RT_1} = \frac{(100 \text{ kPa})(0.0006 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(308 \text{ K})} = 6.788 \times 10^{-4} \text{ kg}$$

$$Q_{\text{in}} = m(u_3 - u_2) = mc_v(T_3 - T_2) = (6.788 \times 10^{-4} \text{ kg})(0.718 \text{ kJ/kg} \cdot \text{K})(1969 - 757.9) \text{ K} = \mathbf{0.590 \text{ kJ}}$$

(c) Process 4-1: $\nu = \text{constant}$ heat rejection.

$$Q_{\text{out}} = m(u_4 - u_1) = mc_v(T_4 - T_1) = -(6.788 \times 10^{-4} \text{ kg})(0.718 \text{ kJ/kg} \cdot \text{K})(800 - 308) \text{ K} = \mathbf{0.240 \text{ kJ}}$$

$$W_{\text{net}} = Q_{\text{in}} - Q_{\text{out}} = 0.590 - 0.240 = 0.350 \text{ kJ}$$

$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_{\text{in}}} = \frac{0.350 \text{ kJ}}{0.590 \text{ kJ}} = \mathbf{59.4\%}$$

$$(d) \quad \nu_{\text{min}} = \nu_2 = \frac{\nu_{\text{max}}}{r}$$

$$\text{MEP} = \frac{W_{\text{net,out}}}{\nu_1 - \nu_2} = \frac{W_{\text{net,out}}}{\nu_1(1 - 1/r)} = \frac{0.350 \text{ kJ}}{(0.0006 \text{ m}^3)(1 - 1/9.5)} \left(\frac{\text{kPa} \cdot \text{m}^3}{\text{kJ}} \right) = \mathbf{652 \text{ kPa}}$$

9-38 Repeat Problem 9-37, but replace the isentropic expansion process by a polytropic expansion process with the polytropic exponent $n = 1.35$.

Assumptions 1 The air-standard assumptions are applicable. 2 Kinetic and potential energy changes are negligible. 3 Air is an ideal gas with constant specific heats.

Properties The properties of air at room temperature are $c_p = 1.005 \text{ kJ/kg}\cdot\text{K}$, $c_v = 0.718 \text{ kJ/kg}\cdot\text{K}$, $R = 0.287 \text{ kJ/kg}\cdot\text{K}$, and $k = 1.4$ (Table A-2).

Analysis (a) Process 1-2: isentropic compression.

$$T_2 = T_1 \left(\frac{v_1}{v_2} \right)^{k-1} = (308 \text{ K})(9.5)^{0.4} = 757.9 \text{ K}$$

$$\frac{P_2 v_2}{T_2} = \frac{P_1 v_1}{T_1} \longrightarrow P_2 = \frac{v_1}{v_2} \frac{T_2}{T_1} P_1 = (9.5) \left(\frac{757.9 \text{ K}}{308 \text{ K}} \right) (100 \text{ kPa}) = 2338 \text{ kPa}$$

Process 3-4: polytropic expansion.

$$m = \frac{P_1 V_1}{RT_1} = \frac{(100 \text{ kPa})(0.0006 \text{ m}^3)}{(0.287 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(308 \text{ K})} = 6.788 \times 10^{-4} \text{ kg}$$

$$T_3 = T_4 \left(\frac{v_4}{v_3} \right)^{n-1} = (800 \text{ K})(9.5)^{0.35} = 1759 \text{ K}$$

$$W_{34} = \frac{mR(T_4 - T_3)}{1 - n} = \frac{(6.788 \times 10^{-4} \text{ kg})(0.287 \text{ kJ/kg}\cdot\text{K})(800 - 1759) \text{ K}}{1 - 1.35} = 0.5338 \text{ kJ}$$

Then energy balance for process 3-4 gives

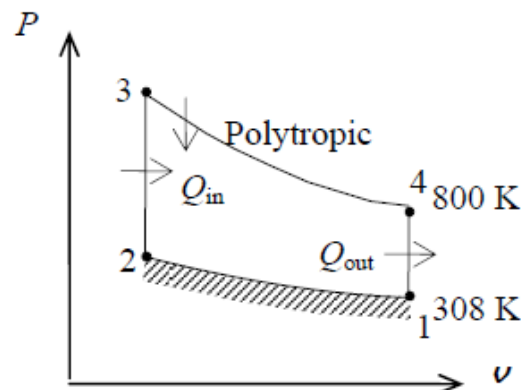
$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

$$Q_{34,\text{in}} - W_{34,\text{out}} = m(u_4 - u_3)$$

$$Q_{34,\text{in}} = m(u_4 - u_3) + W_{34,\text{out}} = mc_v(T_4 - T_3) + W_{34,\text{out}}$$

$$Q_{34,\text{in}} = (6.788 \times 10^{-4} \text{ kg})(0.718 \text{ kJ/kg}\cdot\text{K})(800 - 1759) \text{ K} + 0.5338 \text{ kJ} = 0.0664 \text{ kJ}$$

That is, 0.066 kJ of heat is added to the air during the expansion process (This is not realistic, and probably is due to assuming constant specific heats at room temperature).



$$Pv^n = \text{sabit} \rightarrow W = \int_1^2 P(V) dV = \frac{P_2 V_2 - P_1 V_1}{1 - n}$$

(b) Process 2-3: $\nu = \text{constant}$ heat addition.

$$\frac{P_3 \nu_3}{T_3} = \frac{P_2 \nu_2}{T_2} \longrightarrow P_3 = \frac{T_3}{T_2} P_2 = \left(\frac{1759 \text{ K}}{757.9 \text{ K}} \right) (2338 \text{ kPa}) = \mathbf{5426 \text{ kPa}}$$

$$Q_{23,\text{in}} = m(u_3 - u_2) = mc_\nu(T_3 - T_2)$$

$$Q_{23,\text{in}} = (6.788 \times 10^{-4} \text{ kg})(0.718 \text{ kJ/kg} \cdot \text{K})(1759 - 757.9) \text{ K} = 0.4879 \text{ kJ}$$

Therefore, $Q_{\text{in}} = Q_{23,\text{in}} + Q_{34,\text{in}} = 0.4879 + 0.0664 = \mathbf{0.5543 \text{ kJ}}$

(c) Process 4-1: $\nu = \text{constant}$ heat rejection.

$$Q_{\text{out}} = m(u_4 - u_1) = mc_\nu(T_4 - T_1) = (6.788 \times 10^{-4} \text{ kg})(0.718 \text{ kJ/kg} \cdot \text{K})(800 - 308) \text{ K} = 0.2398 \text{ kJ}$$

$$W_{\text{net,out}} = Q_{\text{in}} - Q_{\text{out}} = 0.5543 - 0.2398 = 0.3145 \text{ kJ}$$

$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_{\text{in}}} = \frac{0.3145 \text{ kJ}}{0.5543 \text{ kJ}} = \mathbf{56.7\%}$$

$$(d) \quad \nu_{\text{min}} = \nu_2 = \frac{\nu_{\text{max}}}{r}$$

$$\text{MEP} = \frac{W_{\text{net,out}}}{\nu_1 - \nu_2} = \frac{W_{\text{net,out}}}{\nu_1(1 - 1/r)} = \frac{0.3145 \text{ kJ}}{(0.0006 \text{ m}^3)(1 - 1/9.5)} \left(\frac{\text{kPa} \cdot \text{m}^3}{\text{kJ}} \right) = \mathbf{586 \text{ kPa}}$$

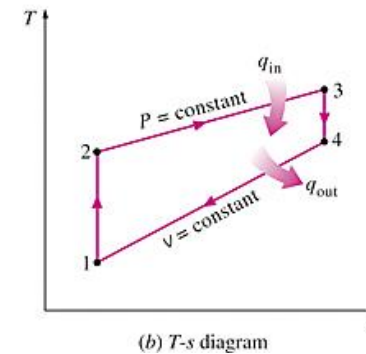
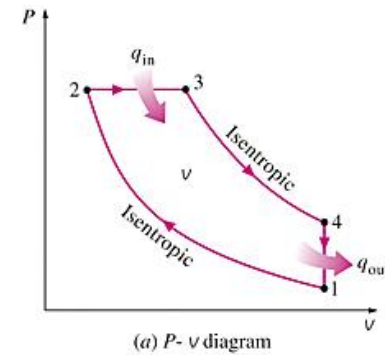
9–41 A four-cylinder, four-stroke, 2.2-L gasoline engine operates on the Otto cycle with a compression ratio of 10. The air is at 100 kPa and 60°C at the beginning of the compression process, and the maximum pressure in the cycle is 8 MPa. The compression and expansion processes may be modeled as polytropic with a polytropic constant of 1.3. Using constant specific heats at 850 K, determine (a) the temperature at the end of the expansion process, (b) the net work output and the thermal efficiency, (c) the mean effective pressure, (d) the engine speed for a net power output of 70 kW, and (e) the specific fuel consumption, in g/kWh, defined as the ratio of the mass of the fuel consumed to the net work produced. The air–fuel ratio, defined as the amount of air divided by the amount of fuel intake, is 16.

DIESEL CYCLE

9-51 An ideal diesel engine has a compression ratio of 20 and uses air as the working fluid. The state of air at the beginning of the compression process is 95 kPa and 20°C. If the maximum temperature in the cycle is not to exceed 2200 K, determine (a) the thermal efficiency and (b) the mean effective pressure. Assume constant specific heats for air at room temperature. *Answers: (a) 63.5 percent, (b) 933 kPa*



Solution



Analysis (a) Process 1-2: isentropic compression.

$$T_2 = T_1 \left(\frac{v_1}{v_2} \right)^{k-1} = (293 \text{ K})(20)^{0.4} = 971.1 \text{ K}$$

Process 2-3: $P = \text{constant}$ heat addition.

$$\frac{P_3 v_3}{T_3} = \frac{P_2 v_2}{T_2} \longrightarrow \frac{v_3}{v_2} = \frac{T_3}{T_2} = \frac{2200 \text{ K}}{971.1 \text{ K}} = 2.265$$

Process 3-4: isentropic expansion.

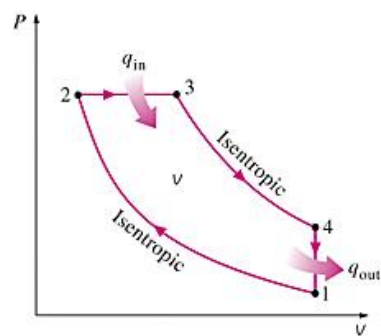
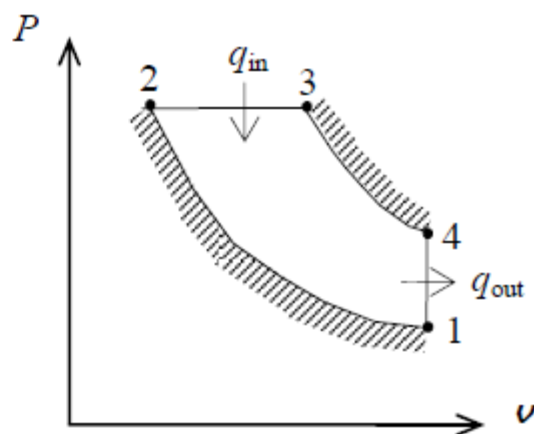
$$T_4 = T_3 \left(\frac{v_3}{v_4} \right)^{k-1} = T_3 \left(\frac{2.265 v_2}{v_4} \right)^{k-1} = T_3 \left(\frac{2.265}{r} \right)^{k-1} = (2200 \text{ K}) \left(\frac{2.265}{20} \right)^{0.4} = 920.6 \text{ K}$$

$$q_{\text{in}} = h_3 - h_2 = c_p (T_3 - T_2) = (1.005 \text{ kJ/kg} \cdot \text{K})(2200 - 971.1) \text{ K} = 1235 \text{ kJ/kg}$$

$$q_{\text{out}} = u_4 - u_1 = c_v (T_4 - T_1) = (0.718 \text{ kJ/kg} \cdot \text{K})(920.6 - 293) \text{ K} = 450.6 \text{ kJ/kg}$$

$$w_{\text{net,out}} = q_{\text{in}} - q_{\text{out}} = 1235 - 450.6 = 784.4 \text{ kJ/kg}$$

$$\eta_{\text{th}} = \frac{w_{\text{net,out}}}{q_{\text{in}}} = \frac{784.4 \text{ kJ/kg}}{1235 \text{ kJ/kg}} = \mathbf{63.5\%}$$

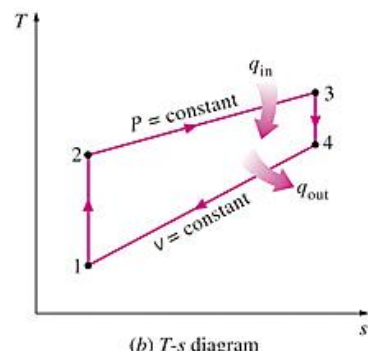


(b)

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(293 \text{ K})}{95 \text{ kPa}} = 0.885 \text{ m}^3/\text{kg} = v_{\text{max}}$$

$$v_{\text{min}} = v_2 = \frac{v_{\text{max}}}{r}$$

$$\text{MEP} = \frac{w_{\text{net,out}}}{v_1 - v_2} = \frac{w_{\text{net,out}}}{v_1 (1 - 1/r)} = \frac{784.4 \text{ kJ/kg}}{(0.885 \text{ m}^3/\text{kg})(1 - 1/20)} \left(\frac{\text{kPa} \cdot \text{m}^3}{\text{kJ}} \right) = \mathbf{933 \text{ kPa}}$$



9-52 Repeat Problem 9-51, but replace the isentropic expansion process by polytropic expansion process with the polytropic exponent $n = 1.35$.

Properties The properties of air at room temperature are $c_p = 1.005 \text{ kJ/kg}\cdot\text{K}$, $c_v = 0.718 \text{ kJ/kg}\cdot\text{K}$, and $k = 1.4$ (Table A-2).

Analysis (a) Process 1-2: isentropic compression.

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{k-1} = (293 \text{ K})(20)^{0.4} = 971.1 \text{ K}$$

Process 2-3: $P = \text{constant}$ heat addition.

$$\frac{P_3 V_3}{T_3} = \frac{P_2 V_2}{T_2} \longrightarrow \frac{V_3}{V_2} = \frac{T_3}{T_2} = \frac{2200 \text{ K}}{971.1 \text{ K}} = 2.265$$

Process 3-4: polytropic expansion.

$$T_4 = T_3 \left(\frac{V_3}{V_4} \right)^{n-1} = T_3 \left(\frac{2.265 V_2}{V_4} \right)^{n-1} = T_3 \left(\frac{2.265}{r} \right)^{n-1} = (2200 \text{ K}) \left(\frac{2.265}{20} \right)^{0.35} = 1026 \text{ K}$$

$$q_{\text{in}} = h_3 - h_2 = c_p (T_3 - T_2) = (1.005 \text{ kJ/kg}\cdot\text{K})(2200 - 971.1) \text{ K} = 1235 \text{ kJ/kg}$$

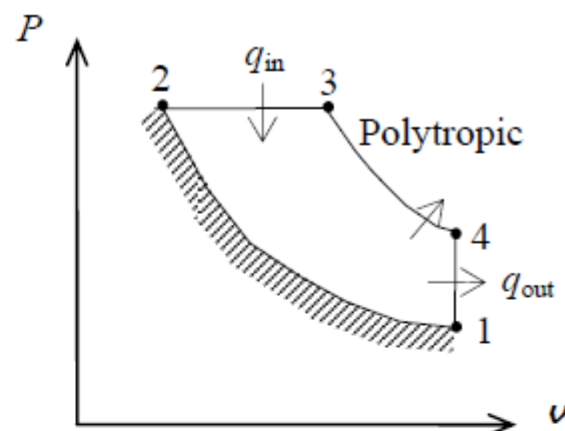
$$q_{\text{out}} = u_4 - u_1 = c_v (T_4 - T_1) = (0.718 \text{ kJ/kg}\cdot\text{K})(1026 - 293) \text{ K} = 526.3 \text{ kJ/kg}$$

Note that q_{out} in this case does not represent the entire heat rejected since some heat is also rejected during the polytropic process, which is determined from an energy balance on process 3-4:

$$w_{34,\text{out}} = \frac{R(T_4 - T_3)}{1 - n} = \frac{(0.287 \text{ kJ/kg}\cdot\text{K})(1026 - 2200) \text{ K}}{1 - 1.35} = 963 \text{ kJ/kg}$$

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

$$\begin{aligned} q_{34,\text{in}} - w_{34,\text{out}} &= u_4 - u_3 \longrightarrow q_{34,\text{in}} = w_{34,\text{out}} + c_v (T_4 - T_3) \\ &= 963 \text{ kJ/kg} + (0.718 \text{ kJ/kg}\cdot\text{K})(1026 - 2200) \text{ K} \\ &= 120.1 \text{ kJ/kg} \end{aligned}$$



which means that 120.1 kJ/kg of heat is transferred to the combustion gases during the expansion process. This is unrealistic since the gas is at a much higher temperature than the surroundings, and a hot gas loses heat during polytropic expansion. The cause of this unrealistic result is the constant specific heat assumption. If we were to use u data from the air table, we would obtain

$$q_{34,\text{in}} = w_{34,\text{out}} + (u_4 - u_3) = 963 + (781.3 - 1872.4) = -128.1 \text{ kJ/kg}$$

which is a heat loss as expected. Then q_{out} becomes

$$q_{\text{out}} = q_{34,\text{out}} + q_{41,\text{out}} = 128.1 + 526.3 = 654.4 \text{ kJ/kg}$$

and

$$w_{\text{net,out}} = q_{\text{in}} - q_{\text{out}} = 1235 - 654.4 = 580.6 \text{ kJ/kg}$$

$$\eta_{\text{th}} = \frac{w_{\text{net,out}}}{q_{\text{in}}} = \frac{580.6 \text{ kJ/kg}}{1235 \text{ kJ/kg}} = 47.0\%$$

$$(b) \quad v_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(293 \text{ K})}{95 \text{ kPa}} = 0.885 \text{ m}^3/\text{kg} = v_{\text{max}}$$

$$v_{\text{min}} = v_2 = \frac{v_{\text{max}}}{r}$$

$$\text{MEP} = \frac{w_{\text{net,out}}}{v_1 - v_2} = \frac{w_{\text{net,out}}}{v_1(1 - 1/r)} = \frac{580.6 \text{ kJ/kg}}{(0.885 \text{ m}^3/\text{kg})(1 - 1/20)} \left(\frac{1 \text{ kPa} \cdot \text{m}^3}{\text{kJ}} \right) = 691 \text{ kPa}$$

9–54 A four-cylinder two-stroke 2.4-L diesel engine that operates on an ideal Diesel cycle has a compression ratio of 17 and a cutoff ratio of 2.2. Air is at 55°C and 97 kPa at the beginning of the compression process. Using the cold-air-standard assumptions, determine how much power the engine will deliver at 1500 rpm.

9–55 Repeat Problem 9–54 using nitrogen as the working fluid.

Stirling and Ericsson Cycles

9-65 Consider an ideal Ericsson cycle with air as the working fluid executed in a steady-flow system. Air is at 27°C and 120 kPa at the beginning of the isothermal compression process, during which 150 kJ/kg of heat is rejected. Heat transfer to air occurs at 1200 K. Determine (a) the maximum pressure in the cycle, (b) the net work output per unit mass of air, and (c) the thermal efficiency of the cycle. *Answers:*

(a) 685 kPa, (b) 450 kJ/kg, (c) 75 percent

Analysis (a) The entropy change during process 3-4 is

$$s_4 - s_3 = -\frac{q_{34,\text{out}}}{T_0} = -\frac{150 \text{ kJ/kg}}{300 \text{ K}} = -0.5 \text{ kJ/kg} \cdot \text{K}$$

and

$$\begin{aligned} s_4 - s_3 &= c_p \ln \frac{T_4}{T_3} - R \ln \frac{P_4}{P_3} \\ &= -(0.287 \text{ kJ/kg} \cdot \text{K}) \ln \frac{P_4}{120 \text{ kPa}} = -0.5 \text{ kJ/kg} \cdot \text{K} \end{aligned}$$

It yields $P_4 = 685.2 \text{ kPa}$

(b) For reversible cycles,

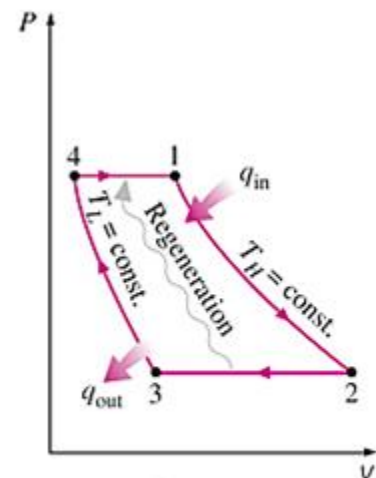
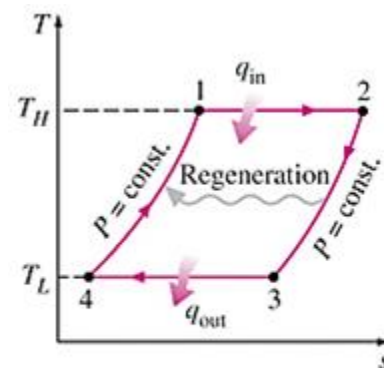
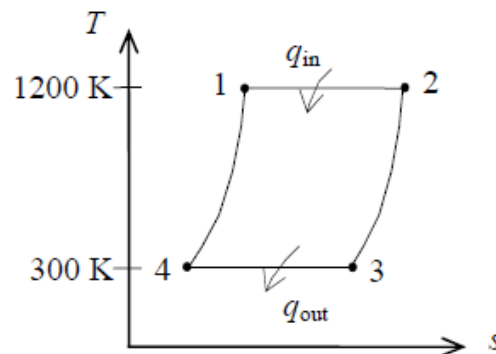
$$\frac{q_{\text{out}}}{q_{\text{in}}} = \frac{T_L}{T_H} \longrightarrow q_{\text{in}} = \frac{T_H}{T_L} q_{\text{out}} = \frac{1200 \text{ K}}{300 \text{ K}} (150 \text{ kJ/kg}) = 600 \text{ kJ/kg}$$

Thus,

$$w_{\text{net,out}} = q_{\text{in}} - q_{\text{out}} = 600 - 150 = 450 \text{ kJ/kg}$$

(c) The thermal efficiency of this totally reversible cycle is determined from

$$\eta_{\text{th}} = 1 - \frac{T_L}{T_H} = 1 - \frac{300 \text{ K}}{1200 \text{ K}} = 75.0\%$$



(c) Ericsson cycle

Ideal and Actual Gas-Turbine (Brayton) Cycles

9-73



A simple Brayton cycle using air as the working fluid has a pressure ratio of 8. The minimum and maximum temperatures in the cycle are 310 and 1160 K. Assuming an isentropic efficiency of 75 percent for the compressor and 82 percent for the turbine, determine (a) the air temperature at the turbine exit, (b) the net work output, and (c) the thermal efficiency.

Assumptions 1 Steady operating conditions exist. 2 The air-standard assumptions are applicable. 3 Kinetic and potential energy changes are negligible. 4 Air is an ideal gas with variable specific heats.

Properties The properties of air are given in Table A-17.

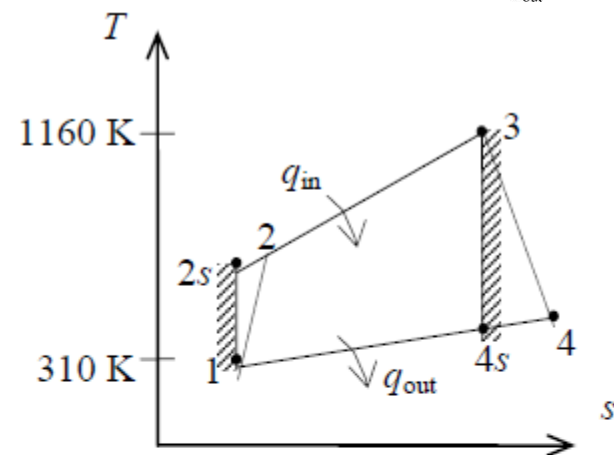
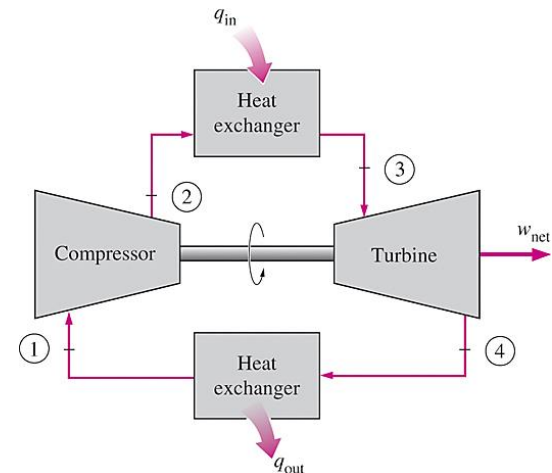
Analysis (a) Noting that process 1-2s is isentropic,

$$T_1 = 310 \text{ K} \longrightarrow \begin{aligned} h_1 &= 310.24 \text{ kJ/kg} \\ P_{r_1} &= 1.5546 \end{aligned}$$

$$P_{r_2} = \frac{P_2}{P_1} P_{r_1} = (8)(1.5546) = 12.44 \longrightarrow h_{2s} = 562.58 \text{ kJ/kg} \text{ and } T_{2s} = 557.25 \text{ K}$$

$$\begin{aligned} \eta_C &= \frac{h_{2s} - h_1}{h_2 - h_1} \longrightarrow h_2 = h_1 + \frac{h_{2s} - h_1}{\eta_C} \\ &= 310.24 + \frac{562.58 - 310.24}{0.75} = 646.7 \text{ kJ/kg} \end{aligned}$$

$$T_3 = 1160 \text{ K} \longrightarrow \begin{aligned} h_3 &= 1230.92 \text{ kJ/kg} \\ P_{r_3} &= 207.2 \end{aligned}$$



$$P_{r_4} = \frac{P_4}{P_3} P_{r_3} = \left(\frac{1}{8}\right)(207.2) = 25.90 \longrightarrow h_{4s} = 692.19 \text{ kJ/kg and } T_{4s} = 680.3 \text{ K}$$

$$\begin{aligned} \eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}} &\longrightarrow h_4 = h_3 - \eta_T (h_3 - h_{4s}) \\ &= 1230.92 - (0.82)(1230.92 - 692.19) \\ &= 789.16 \text{ kJ/kg} \end{aligned}$$

Thus,

$$T_4 = 770.1 \text{ K}$$

$$(b) \quad q_{\text{in}} = h_3 - h_2 = 1230.92 - 646.7 = 584.2 \text{ kJ/kg}$$

$$q_{\text{out}} = h_4 - h_1 = 789.16 - 310.24 = 478.92 \text{ kJ/kg}$$

$$w_{\text{net,out}} = q_{\text{in}} - q_{\text{out}} = 584.2 - 478.92 = \mathbf{105.3 \text{ kJ/kg}}$$

$$(c) \quad \eta_{\text{th}} = \frac{w_{\text{net,out}}}{q_{\text{in}}} = \frac{105.3 \text{ kJ/kg}}{584.2 \text{ kJ/kg}} = \mathbf{18.0\%}$$

9-75 Repeat Problem 9-73 using constant specific heats at room temperature.

Analysis (a) Using the compressor and turbine efficiency relations,

$$T_{2s} = T_1 \left(\frac{P_2}{P_1} \right)^{(k-1)/k} = (310 \text{ K})(8)^{0.4/1.4} = 561.5 \text{ K}$$

$$T_{4s} = T_3 \left(\frac{P_4}{P_3} \right)^{(k-1)/k} = (1160 \text{ K}) \left(\frac{1}{8} \right)^{0.4/1.4} = 640.4 \text{ K}$$

$$\eta_C = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{c_p (T_{2s} - T_1)}{c_p (T_2 - T_1)} \longrightarrow T_2 = T_1 + \frac{T_{2s} - T_1}{\eta_C}$$

$$= 310 + \frac{561.5 - 310}{0.75} = 645.3 \text{ K}$$

$$\eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}} = \frac{c_p (T_3 - T_4)}{c_p (T_3 - T_{4s})} \longrightarrow T_4 = T_3 - \eta_T (T_3 - T_{4s})$$

$$= 1160 - (0.82)(1160 - 640.4)$$

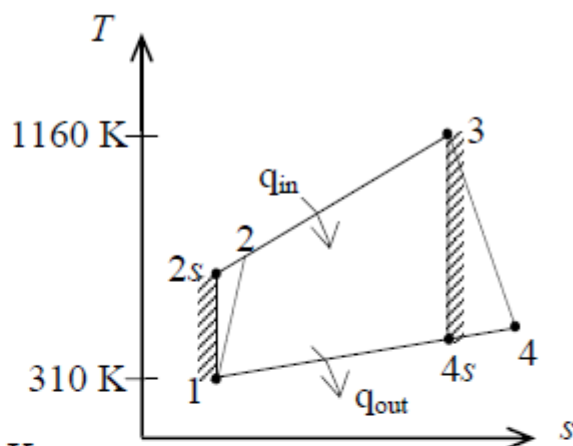
$$= \mathbf{733.9 \text{ K}}$$

$$(b) \quad q_{\text{in}} = h_3 - h_2 = c_p (T_3 - T_2) = (1.005 \text{ kJ/kg} \cdot \text{K})(1160 - 645.3) \text{ K} = 517.3 \text{ kJ/kg}$$

$$q_{\text{out}} = h_4 - h_1 = c_p (T_4 - T_1) = (1.005 \text{ kJ/kg} \cdot \text{K})(733.9 - 310) \text{ K} = 426.0 \text{ kJ/kg}$$

$$w_{\text{net,out}} = q_{\text{in}} - q_{\text{out}} = 517.3 - 426.0 = \mathbf{91.3 \text{ kJ/kg}}$$

$$(c) \quad \eta_{\text{th}} = \frac{w_{\text{net,out}}}{q_{\text{in}}} = \frac{91.3 \text{ kJ/kg}}{517.3 \text{ kJ/kg}} = \mathbf{17.6\%}$$



9-76 Air is used as the working fluid in a simple ideal Brayton cycle that has a pressure ratio of 12, a compressor inlet temperature of 300 K, and a turbine inlet temperature of 1000 K. Determine the required mass flow rate of air for a net power output of 70 MW, assuming both the compressor and the turbine have an isentropic efficiency of (a) 100 percent and (b) 85 percent. Assume constant specific heats at room temperature. *Answers: (a) 352 kg/s, (b) 1037 kg/s*

9-77 A stationary gas-turbine power plant operates on a simple ideal Brayton cycle with air as the working fluid. The air enters the compressor at 95 kPa and 290 K and the turbine at 760 kPa and 1100 K. Heat is transferred to air at a rate of 35,000 kJ/s. Determine the power delivered by this plant (a) assuming constant specific heats at room temperature and (b) accounting for the variation of specific heats with temperature.



Solution

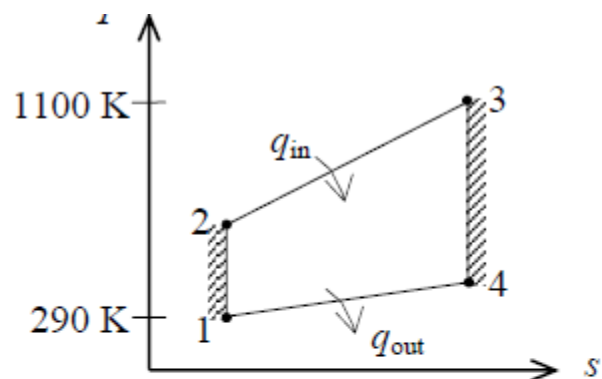
Analysis (a) Assuming constant specific heats,

$$T_{2s} = T_1 \left(\frac{P_2}{P_1} \right)^{(k-1)/k} = (290 \text{ K})(8)^{0.4/1.4} = 525.3 \text{ K}$$

$$T_{4s} = T_3 \left(\frac{P_4}{P_3} \right)^{(k-1)/k} = (1100 \text{ K}) \left(\frac{1}{8} \right)^{0.4/1.4} = 607.2 \text{ K}$$

$$\eta_{\text{th}} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{c_p (T_4 - T_1)}{c_p (T_3 - T_2)} = 1 - \frac{T_4 - T_1}{T_3 - T_2} = 1 - \frac{607.2 - 290}{1100 - 525.3} = 0.448$$

$$\dot{W}_{\text{net,out}} = \eta_{\text{th}} \dot{Q}_{\text{in}} = (0.448)(35,000 \text{ kW}) = \mathbf{15,680 \text{ kW}}$$



(b) Assuming variable specific heats (Table A-17),

$$T_1 = 290 \text{ K} \longrightarrow \begin{aligned} h_1 &= 290.16 \text{ kJ/kg} \\ P_{r_1} &= 1.2311 \end{aligned}$$

$$P_{r_2} = \frac{P_2}{P_1} P_{r_1} = (8)(1.2311) = 9.8488 \longrightarrow h_2 = 526.12 \text{ kJ/kg}$$

$$T_3 = 1100 \text{ K} \longrightarrow \begin{aligned} h_3 &= 1161.07 \text{ kJ/kg} \\ P_{r_3} &= 167.1 \end{aligned}$$

$$P_{r_4} = \frac{P_4}{P_3} P_{r_3} = \left(\frac{1}{8} \right) (167.1) = 20.89 \longrightarrow h_4 = 651.37 \text{ kJ/kg}$$

$$\eta_{\text{th}} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{h_4 - h_1}{h_3 - h_2} = 1 - \frac{651.37 - 290.16}{1161.07 - 526.11} = 0.431$$

$$\dot{W}_{\text{net,out}} = \eta_T \dot{Q}_{\text{in}} = (0.431)(35,000 \text{ kW}) = \mathbf{15,085 \text{ kW}}$$

9–78 Air enters the compressor of a gas-turbine engine at 300 K and 100 kPa, where it is compressed to 700 kPa and 580 K. Heat is transferred to air in the amount of 950 kJ/kg before it enters the turbine. For a turbine efficiency of 86 percent, determine (a) the fraction of the turbine work output used to drive the compressor and (b) the thermal efficiency. Assume variable specific heats for air.

9–79 Repeat Problem 9–78 using constant specific heats at room temperature.

9–84 A gas-turbine power plant operates on the simple Brayton cycle between the pressure limits of 100 and 1200 kPa. The working fluid is air, which enters the compressor at 30°C at a rate of 150 m³/min and leaves the turbine at 500°C. Using variable specific heats for air and assuming a compressor isentropic efficiency of 82 percent and a turbine isentropic efficiency of 88 percent, determine (a) the net power output, (b) the back work ratio, and (c) the thermal efficiency.

Answers: (a) 659 kW, (b) 0.625, (c) 0.319

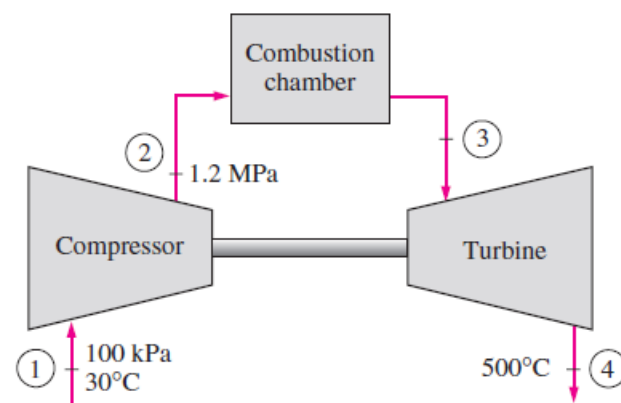


FIGURE P9–84

Analysis (a) For this problem, we use the properties from EES software. Remember that for an ideal gas, enthalpy is a function of temperature only whereas entropy is functions of both temperature and pressure.

Process 1-2: Compression

$$T_1 = 30^\circ\text{C} \longrightarrow h_1 = 303.60 \text{ kJ/kg}$$

$$\left. \begin{array}{l} T_1 = 30^\circ\text{C} \\ P_1 = 100 \text{ kPa} \end{array} \right\} s_1 = 5.7159 \text{ kJ/kg} \cdot \text{K}$$

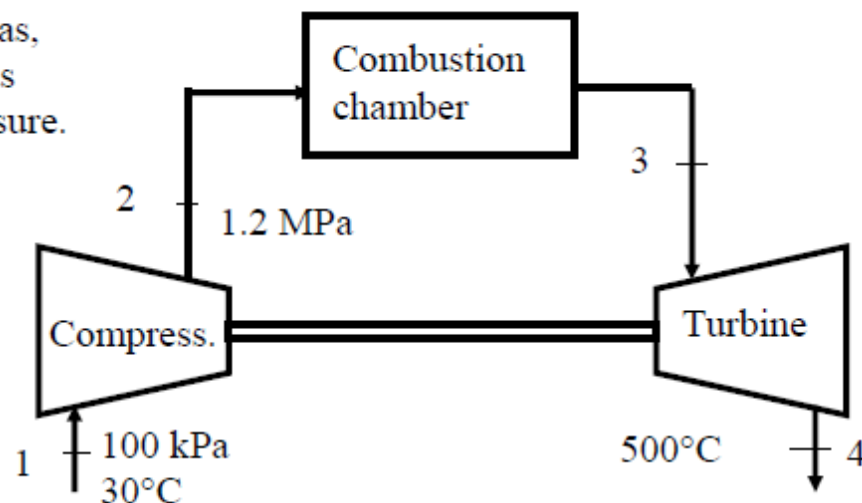
$$\left. \begin{array}{l} P_2 = 1200 \text{ kPa} \\ s_2 = s_1 = 5.7159 \text{ kJ/kg} \cdot \text{K} \end{array} \right\} h_{2s} = 617.37 \text{ kJ/kg}$$

$$\eta_C = \frac{h_{2s} - h_1}{h_2 - h_1} \longrightarrow 0.82 = \frac{617.37 - 303.60}{h_2 - 303.60} \longrightarrow h_2 = 686.24 \text{ kJ/kg}$$

Process 3-4: Expansion

$$T_4 = 500^\circ\text{C} \longrightarrow h_4 = 792.62 \text{ kJ/kg}$$

$$\eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}} \longrightarrow 0.88 = \frac{h_3 - 792.62}{h_3 - h_{4s}}$$



We cannot find the enthalpy at state 3 directly. However, using the following lines in EES together with the isentropic efficiency relation, we find $h_3 = 1404.7$ kJ/kg, $T_3 = 1034^\circ\text{C}$, $s_3 = 6.5699$ kJ/kg.K. The solution by hand would require a trial-error approach.

$$h_3 = \text{enthalpy}(\text{Air}, T=T_3)$$

$$s_3 = \text{entropy}(\text{Air}, T=T_3, P=P_2)$$

$$h_{4s} = \text{enthalpy}(\text{Air}, P=P_1, s=s_3)$$

The mass flow rate is determined from

$$\dot{m} = \frac{P_1 \dot{V}_1}{RT_1} = \frac{(100 \text{ kPa})(150/60 \text{ m}^3/\text{s})}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(30 + 273 \text{ K})} = 2.875 \text{ kg/s}$$

The net power output is

$$\dot{W}_{C,\text{in}} = \dot{m}(h_2 - h_1) = (2.875 \text{ kg/s})(686.24 - 303.60) \text{ kJ/kg} = 1100 \text{ kW}$$

$$\dot{W}_{T,\text{out}} = \dot{m}(h_3 - h_4) = (2.875 \text{ kg/s})(1404.7 - 792.62) \text{ kJ/kg} = 1759 \text{ kW}$$

$$\dot{W}_{\text{net}} = \dot{W}_{T,\text{out}} - \dot{W}_{C,\text{in}} = 1759 - 1100 = \mathbf{659 \text{ kW}}$$

(b) The back work ratio is

$$r_{\text{bw}} = \frac{\dot{W}_{C,\text{in}}}{\dot{W}_{T,\text{out}}} = \frac{1100 \text{ kW}}{1759 \text{ kW}} = \mathbf{0.625}$$

(c) The rate of heat input and the thermal efficiency are

$$\dot{Q}_{\text{in}} = \dot{m}(h_3 - h_2) = (2.875 \text{ kg/s})(1404.7 - 686.24) \text{ kJ/kg} = 2065 \text{ kW}$$

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net}}}{\dot{Q}_{\text{in}}} = \frac{659 \text{ kW}}{2065 \text{ kW}} = \mathbf{0.319}$$

9-91



The 7FA gas turbine manufactured by General Electric is reported to have an efficiency of 35.9 percent in the simple-cycle mode and to produce 159 MW of net power. The pressure ratio is 14.7 and the turbine inlet temperature is 1288°C. The mass flow rate through the turbine is 1,536,000 kg/h. Taking the ambient conditions to be 20°C and 100 kPa, determine the isentropic efficiency of the turbine and the compressor. Also, determine the thermal efficiency of this gas turbine if a regenerator with an effectiveness of 80 percent is added.

Properties The properties of air are given in Table A-17.

Analysis The properties at various states are

$$T_1 = 20^\circ\text{C} = 293 \text{ K} \longrightarrow \begin{aligned} h_1 &= 293.2 \text{ kJ/kg} \\ P_{r_1} &= 1.2765 \end{aligned}$$

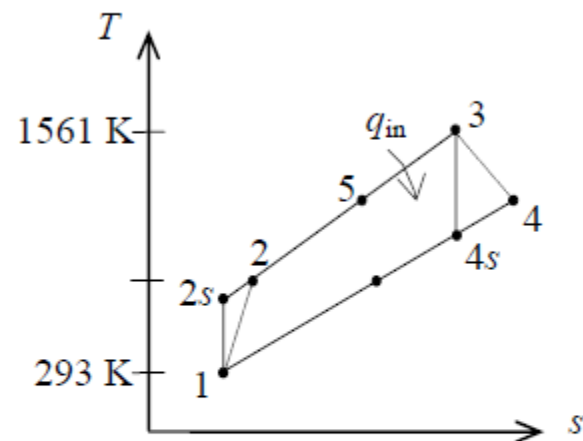
$$P_{r_2} = \frac{P_2}{P_1} P_{r_1} = (14.7)(1.2765) = 18.765 \longrightarrow h_{2s} = 643.3 \text{ kJ/kg}$$

$$T_3 = 1288^\circ\text{C} = 1561 \text{ K} \longrightarrow \begin{aligned} h_3 &= 1710.0 \text{ kJ/kg} \\ P_{r_3} &= 712.5 \end{aligned}$$

$$P_{r_4} = \frac{P_4}{P_3} P_{r_3} = \left(\frac{1}{14.7} \right) (712.5) = 48.47 \longrightarrow h_{4s} = 825.23 \text{ kJ/kg}$$

The net work output and the heat input per unit mass are

$$w_{\text{net}} = \frac{\dot{W}_{\text{net}}}{\dot{m}} = \frac{159,000 \text{ kW}}{1,536,000 \text{ kg/h}} \left(\frac{3600 \text{ s}}{1 \text{ h}} \right) = 372.66 \text{ kJ/kg}$$



$$q_{\text{in}} = \frac{w_{\text{net}}}{\eta_{\text{th}}} = \frac{372.66 \text{ kJ/kg}}{0.359} = 1038.0 \text{ kJ/kg}$$

$$q_{\text{in}} = h_3 - h_2 \rightarrow h_2 = h_3 - q_{\text{in}} = 1710 - 1038 = 672.0 \text{ kJ/kg}$$

$$q_{\text{out}} = q_{\text{in}} - w_{\text{net}} = 1038.0 - 372.66 = 665.34 \text{ kJ/kg}$$

$$q_{\text{out}} = h_4 - h_1 \rightarrow h_4 = q_{\text{out}} + h_1 = 665.34 + 293.2 = 958.54 \text{ kJ/kg} \rightarrow T_4 = 650^\circ\text{C}$$

Then the compressor and turbine efficiencies become

$$\eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}} = \frac{1710 - 958.54}{1710 - 825.23} = \mathbf{0.849}$$

$$\eta_C = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{643.3 - 293.2}{672 - 293.2} = \mathbf{0.924}$$

When a regenerator is added, the new heat input and the thermal efficiency become

$$q_{\text{regen}} = \varepsilon(h_4 - h_2) = (0.80)(958.54 - 672.0) = 286.54 \text{ kJ/kg}$$

$$q_{\text{in,new}} = q_{\text{in}} - q_{\text{regen}} = 1038 - 286.54 = 751.46 \text{ kJ/kg}$$

$$\eta_{\text{th,new}} = \frac{w_{\text{net}}}{q_{\text{in,new}}} = \frac{372.66 \text{ kJ/kg}}{751.46 \text{ kJ/kg}} = \mathbf{0.496}$$

Discussion Note an 80% efficient regenerator would increase the thermal efficiency of this gas turbine from 35.9% to 49.6%.

9-96 A Brayton cycle with regeneration using air as the working fluid has a pressure ratio of 7. The minimum and maximum temperatures in the cycle are 310 and 1150 K. Assuming an isentropic efficiency of 75 percent for the compressor and 82 percent for the turbine and an effectiveness of 65 percent for the regenerator, determine (a) the air temperature at the turbine exit, (b) the net work output, and (c) the thermal efficiency.

Answers: (a) 783 K, (b) 108.1 kJ/kg, (c) 22.5 percent

9-97 A stationary gas-turbine power plant operates on an ideal regenerative Brayton cycle ($\epsilon = 100$ percent) with air as the working fluid. Air enters the compressor at 95 kPa and 290 K and the turbine at 760 kPa and 1100 K. Heat is transferred to air from an external source at a rate of 75,000 kJ/s. Determine the power delivered by this plant (a) assuming constant specific heats for air at room temperature and (b) accounting for the variation of specific heats with temperature.

Analysis (a) Assuming constant specific heats,

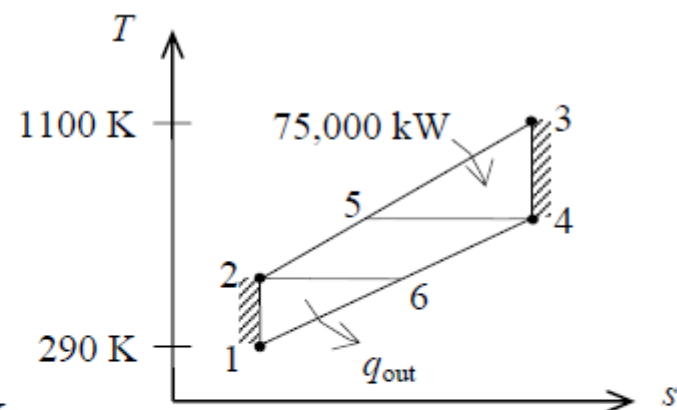
$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{(k-1)/k} = (290 \text{ K})(8)^{0.4/1.4} = 525.3 \text{ K}$$

$$T_4 = T_3 \left(\frac{P_4}{P_3} \right)^{(k-1)/k} = (1100 \text{ K}) \left(\frac{1}{8} \right)^{0.4/1.4} = 607.2 \text{ K}$$

$$\epsilon = 100\% \longrightarrow T_5 = T_4 = 607.2 \text{ K and } T_6 = T_2 = 525.3 \text{ K}$$

$$\eta_{\text{th}} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{c_p(T_6 - T_1)}{c_p(T_3 - T_5)} = 1 - \frac{T_6 - T_1}{T_3 - T_5} = 1 - \frac{525.3 - 290}{1100 - 607.2} = 0.5225$$

$$\dot{W}_{\text{net}} = \eta_T \dot{Q}_{\text{in}} = (0.5225)(75,000 \text{ kW}) = \mathbf{39,188 \text{ kW}}$$



(b) Assuming variable specific heats,

$$T_1 = 290\text{K} \longrightarrow \begin{matrix} h_1 = 290.16 \text{ kJ/kg} \\ P_{r_1} = 1.2311 \end{matrix}$$

$$P_{r_2} = \frac{P_2}{P_1} P_{r_1} = (8)(1.2311) = 9.8488 \longrightarrow h_2 = 526.12 \text{ kJ/kg}$$

$$T_3 = 1100\text{K} \longrightarrow \begin{matrix} h_3 = 1161.07 \text{ kJ/kg} \\ P_{r_3} = 167.1 \end{matrix}$$

$$P_{r_4} = \frac{P_4}{P_3} P_{r_3} = \left(\frac{1}{8}\right)(167.1) = 20.89 \longrightarrow h_4 = 651.37 \text{ kJ/kg}$$

$$\varepsilon = 100\% \longrightarrow h_5 = h_4 = 651.37 \text{ kJ/kg and } h_6 = h_2 = 526.12 \text{ kJ/kg}$$

$$\eta_{\text{th}} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{h_6 - h_1}{h_3 - h_5} = 1 - \frac{526.12 - 290.16}{1161.07 - 651.37} = 0.5371$$

$$\dot{W}_{\text{net}} = \eta_T \dot{Q}_{\text{in}} = (0.5371)(75,000 \text{ kW}) = \mathbf{40,283 \text{ kW}}$$

9-98 Air enters the compressor of a regenerative gas-turbine engine at 300 K and 100 kPa, where it is compressed to 800 kPa and 580 K. The regenerator has an effectiveness of 72 percent, and the air enters the turbine at 1200 K. For a turbine efficiency of 86 percent, determine (a) the amount of heat transfer in the regenerator and (b) the thermal efficiency. Assume variable specific heats for air. *Answers: (a) 152.5 kJ/kg, (b) 36.0 percent*

9-99 Repeat Problem 9-98 using constant specific heats at room temperature.

9-100 Repeat Problem 9-98 for a regenerator effectiveness of 70 percent.

Brayton Cycle with Intercooling, Reheating, and Regeneration

9-108 Consider an ideal gas-turbine cycle with two stages of compression and two stages of expansion. The pressure ratio across each stage of the compressor and turbine is 3. The air enters each stage of the compressor at 300 K and each stage of the turbine at 1200 K. Determine the back work ratio and the thermal efficiency of the cycle, assuming (a) no regenerator is used and (b) a regenerator with 75 percent effectiveness is used. Use variable specific heats.

Assumptions 1 The air standard assumptions are applicable. **2** Air is an ideal gas with variable specific heats. **3** Kinetic and potential energy changes are negligible.

Properties The properties of air are given in Table A-17.

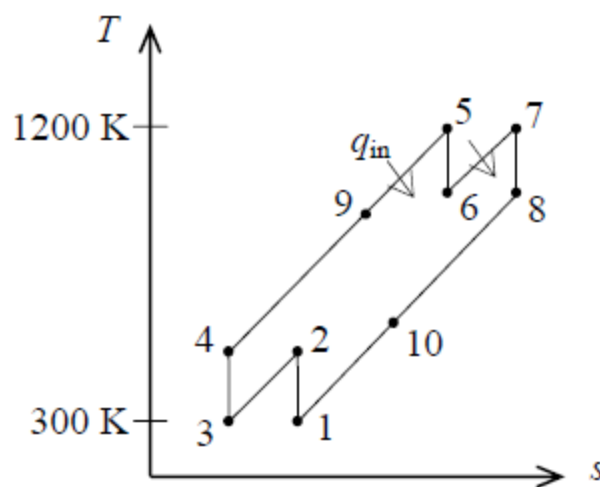
Analysis (a) The work inputs to each stage of compressor are identical, so are the work outputs of each stage of the turbine since this is an ideal cycle. Then,

$$T_1 = 300 \text{ K} \longrightarrow \begin{aligned} h_1 &= 300.19 \text{ kJ/kg} \\ P_{r_1} &= 1.386 \end{aligned}$$

$$P_{r_2} = \frac{P_2}{P_1} P_{r_1} = (3)(1.386) = 4.158 \longrightarrow h_2 = h_4 = 411.26 \text{ kJ/kg}$$

$$T_5 = 1200 \text{ K} \longrightarrow \begin{aligned} h_5 &= h_7 = 1277.79 \text{ kJ/kg} \\ P_{r_5} &= 238 \end{aligned}$$

$$P_{r_6} = \frac{P_6}{P_5} P_{r_5} = \left(\frac{1}{3}\right)(238) = 79.33 \longrightarrow h_6 = h_8 = 946.36 \text{ kJ/kg}$$



$$w_{C,in} = 2(h_2 - h_1) = 2(411.26 - 300.19) = 222.14 \text{ kJ/kg}$$

$$w_{T,out} = 2(h_5 - h_6) = 2(1277.79 - 946.36) = 662.86 \text{ kJ/kg}$$

Thus,

$$r_{bw} = \frac{w_{C,in}}{w_{T,out}} = \frac{222.14 \text{ kJ/kg}}{662.86 \text{ kJ/kg}} = \mathbf{33.5\%}$$

$$q_{in} = (h_5 - h_4) + (h_7 - h_6) = (1277.79 - 411.26) + (1277.79 - 946.36) = 1197.96 \text{ kJ/kg}$$

$$w_{net} = w_{T,out} - w_{C,in} = 662.86 - 222.14 = 440.72 \text{ kJ/kg}$$

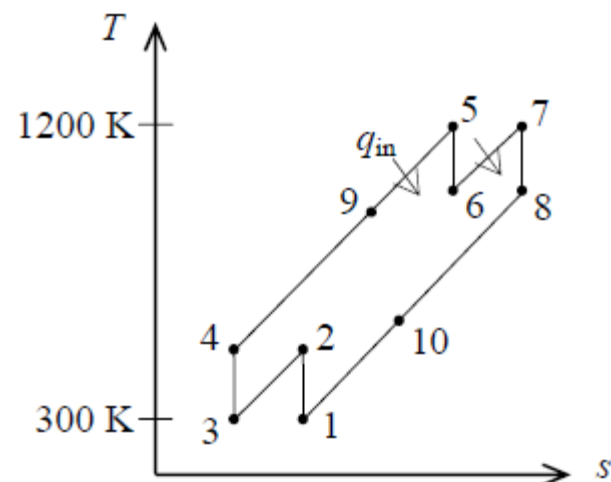
$$\eta_{th} = \frac{w_{net}}{q_{in}} = \frac{440.72 \text{ kJ/kg}}{1197.96 \text{ kJ/kg}} = \mathbf{36.8\%}$$

(b) When a regenerator is used, r_{bw} remains the same. The thermal efficiency in this case becomes

$$q_{regen} = \varepsilon(h_8 - h_4) = (0.75)(946.36 - 411.26) = 401.33 \text{ kJ/kg}$$

$$q_{in} = q_{in,old} - q_{regen} = 1197.96 - 401.33 = 796.63 \text{ kJ/kg}$$

$$\eta_{th} = \frac{w_{net}}{q_{in}} = \frac{440.72 \text{ kJ/kg}}{796.63 \text{ kJ/kg}} = \mathbf{55.3\%}$$



9–109 Repeat Problem 9–108, assuming an efficiency of 80 percent for each compressor stage and an efficiency of 85 percent for each turbine stage.

9-117 A turbojet aircraft is flying with a velocity of 320 m/s at an altitude of 9150 m, where the ambient conditions are 32 kPa and -32°C . The pressure ratio across the compressor is 12, and the temperature at the turbine inlet is 1400 K. Air enters the compressor at a rate of 60 kg/s, and the jet fuel has a heating value of 42,700 kJ/kg. Assuming ideal operation for all components and constant specific heats for air at room temperature, determine (a) the velocity of the exhaust gases, (b) the propulsive power developed, and (c) the rate of fuel consumption.



Solution

Properties The properties of air at room temperature are $c_p = 1.005$ kJ/kg.K and $k = 1.4$ (Table A-2a).

Analysis (a) We assume the aircraft is stationary and the air is moving towards the aircraft at a velocity of $V_1 = 320$ m/s. Ideally, the air will leave the diffuser with a negligible velocity ($V_2 \cong 0$).

Diffuser:

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \quad \text{No (steady)} \longrightarrow \dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$h_1 + V_1^2 / 2 = h_2 + V_2^2 / 2 \longrightarrow 0 = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2}$$

$$0 = c_p (T_2 - T_1) - V_1^2 / 2$$

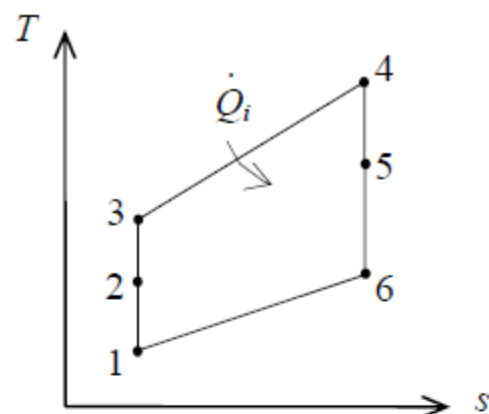
$$T_2 = T_1 + \frac{V_1^2}{2c_p} = 241 \text{ K} + \frac{(320 \text{ m/s})^2}{(2)(1.005 \text{ kJ/kg} \cdot \text{K})} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = 291.9 \text{ K}$$

$$P_2 = P_1 \left(\frac{T_2}{T_1} \right)^{k/(k-1)} = (32 \text{ kPa}) \left(\frac{291.9 \text{ K}}{241 \text{ K}} \right)^{1.4/0.4} = 62.6 \text{ kPa}$$

Compressor:

$$P_3 = P_4 = (r_p)(P_2) = (12)(62.6 \text{ kPa}) = 751.2 \text{ kPa}$$

$$T_3 = T_2 \left(\frac{P_3}{P_2} \right)^{(k-1)/k} = (291.9 \text{ K})(12)^{0.4/1.4} = 593.7 \text{ K}$$



$$w_{\text{comp,in}} = w_{\text{turb,out}} \longrightarrow h_3 - h_2 = h_4 - h_5 \longrightarrow c_p (T_3 - T_2) = c_p (T_4 - T_5)$$

or

$$T_5 = T_4 - T_3 + T_2 = 1400 - 593.7 + 291.9 = 1098.2 \text{ K}$$

Nozzle:

$$T_6 = T_4 \left(\frac{P_6}{P_4} \right)^{(k-1)/k} = (1400 \text{ K}) \left(\frac{32 \text{ kPa}}{751.2 \text{ kPa}} \right)^{0.4/1.4} = 568.2 \text{ K}$$

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \Delta \dot{E}_{\text{system}} \overset{\text{0 (steady)}}{\longrightarrow} \dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$h_5 + V_5^2 / 2 = h_6 + V_6^2 / 2$$

$$0 = h_6 - h_5 + \frac{V_6^2 - V_5^2}{2} \longrightarrow 0 = c_p (T_6 - T_5) + V_6^2 / 2$$

$$V_6 = \sqrt{(2)(1.005 \text{ kJ/kg} \cdot \text{K})(1098.2 - 568.2) \text{ K} \left(\frac{1000 \text{ m}^2/\text{s}^2}{1 \text{ kJ/kg}} \right)} = 1032 \text{ m/s}$$

$$(b) \quad \dot{W}_p = \dot{m}(V_{\text{exit}} - V_{\text{inlet}})V_{\text{aircraft}} = (60 \text{ kg/s})(1032 - 320) \text{ m/s}(320 \text{ m/s}) \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = 13,670 \text{ kW}$$

$$(c) \quad \dot{Q}_{\text{in}} = \dot{m}(h_4 - h_3) = \dot{m}c_p (T_4 - T_3) = (60 \text{ kg/s})(1.005 \text{ kJ/kg} \cdot \text{K})(1400 - 593.7) \text{ K} = 48,620 \text{ kJ/s}$$

$$\dot{m}_{\text{fuel}} = \frac{\dot{Q}_{\text{in}}}{\text{HV}} = \frac{48,620 \text{ kJ/s}}{42,700 \text{ kJ/kg}} = 1.14 \text{ kg/s}$$

9-118 Repeat Problem 9-117 using a compressor efficiency of 80 percent and a turbine efficiency of 85 percent.

9-121 Air at 7°C enters a turbojet engine at a rate of 16 kg/s and at a velocity of 300 m/s (relative to the engine). Air is heated in the combustion chamber at a rate 15,000 kJ/s and it leaves the engine at 427°C . Determine the thrust produced by this turbojet engine. (*Hint:* Choose the entire engine as your control volume.)

Second-Law Analysis of Gas Power Cycles

9-129 A gas-turbine power plant operates on the simple Brayton cycle between the pressure limits of 100 and 700 kPa. Air enters the compressor at 30°C at a rate of 12.6 kg/s and leaves at 260°C. A diesel fuel with a heating value of 42,000 kJ/kg is burned in the combustion chamber with an air–fuel ratio of 60 and a combustion efficiency of 97 percent. Combustion gases leave the combustion chamber and enter the turbine whose isentropic efficiency is 85 percent. Treating the combustion gases as air and using constant specific heats at 500°C, determine (a) the isentropic efficiency of the compressor, (b) the net power output and the back work ratio, (c) the thermal efficiency, and (d) the second-law efficiency.

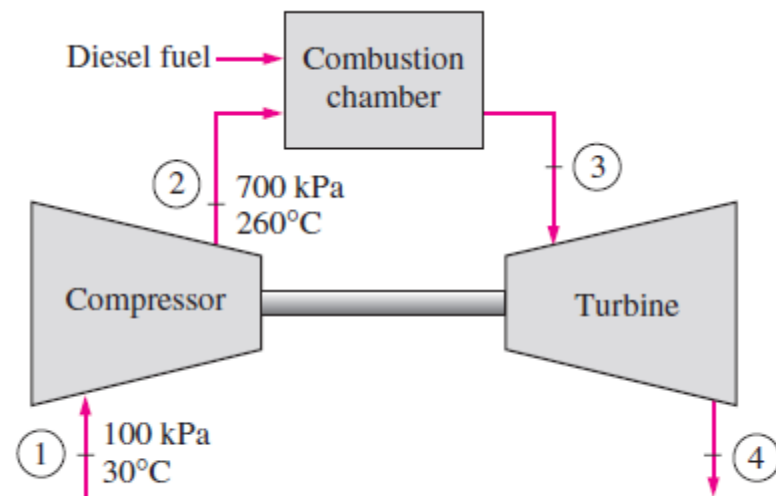


FIGURE P9-129

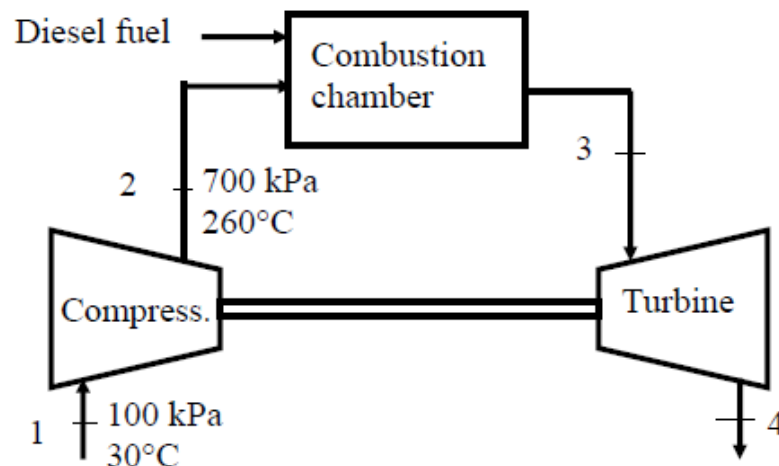
Assumptions 1 The air-standard assumptions are applicable. 2 Kinetic and potential energy changes are negligible. 3 Air is an ideal gas with constant specific heats.

Properties The properties of air at 500°C = 773 K are $c_p = 1.093$ kJ/kg·K, $c_v = 0.806$ kJ/kg·K, $R = 0.287$ kJ/kg·K, and $k = 1.357$ (Table A-2b).

Analysis (a) The isentropic efficiency of the compressor may be determined if we first calculate the exit temperature for the isentropic case

$$T_{2s} = T_1 \left(\frac{P_2}{P_1} \right)^{(k-1)/k} = (303 \text{ K}) \left(\frac{700 \text{ kPa}}{100 \text{ kPa}} \right)^{(1.357-1)/1.357} = 505.6 \text{ K}$$

$$\eta_c = \frac{T_{2s} - T_1}{T_2 - T_1} = \frac{(505.6 - 303) \text{ K}}{(533 - 303) \text{ K}} = \mathbf{0.881}$$



(b) The total mass flowing through the turbine and the rate of heat input are

$$\dot{m}_t = \dot{m}_a + \dot{m}_f = \dot{m}_a + \frac{\dot{m}_a}{\text{AF}} = 12.6 \text{ kg/s} + \frac{12.6 \text{ kg/s}}{60} = 12.6 \text{ kg/s} + 0.21 \text{ kg/s} = 12.81 \text{ kg/s}$$

$$\dot{Q}_{\text{in}} = \dot{m}_f q_{\text{HV}} \eta_c = (0.21 \text{ kg/s})(42,000 \text{ kJ/kg})(0.97) = 8555 \text{ kW}$$

The temperature at the exit of combustion chamber is

$$\dot{Q}_{\text{in}} = \dot{m} c_p (T_3 - T_2) \longrightarrow 8555 \text{ kJ/s} = (12.81 \text{ kg/s})(1.093 \text{ kJ/kg}\cdot\text{K})(T_3 - 533)\text{K} \longrightarrow T_3 = 1144 \text{ K}$$

The temperature at the turbine exit is determined using isentropic efficiency relation

$$T_{4s} = T_3 \left(\frac{P_4}{P_3} \right)^{(k-1)/k} = (1144 \text{ K}) \left(\frac{100 \text{ kPa}}{700 \text{ kPa}} \right)^{(1.357-1)/1.357} = 685.7 \text{ K}$$

$$\eta_T = \frac{T_3 - T_4}{T_3 - T_{4s}} \longrightarrow 0.85 = \frac{(1144 - T_4)\text{K}}{(1144 - 685.7)\text{K}} \longrightarrow T_4 = 754.4 \text{ K}$$

The net power and the back work ratio are

$$\dot{W}_{\text{C,in}} = \dot{m}_a c_p (T_2 - T_1) = (12.6 \text{ kg/s})(1.093 \text{ kJ/kg}\cdot\text{K})(533 - 303)\text{K} = 3168 \text{ kW}$$

$$\dot{W}_{\text{T,out}} = \dot{m} c_p (T_3 - T_4) = (12.81 \text{ kg/s})(1.093 \text{ kJ/kg}\cdot\text{K})(1144 - 754.4)\text{K} = 5455 \text{ kW}$$

$$\dot{W}_{\text{net}} = \dot{W}_{\text{T,out}} - \dot{W}_{\text{C,in}} = 5455 - 3168 = \mathbf{2287 \text{ kW}}$$

$$r_{\text{bw}} = \frac{\dot{W}_{\text{C,in}}}{\dot{W}_{\text{T,out}}} = \frac{3168 \text{ kW}}{5455 \text{ kW}} = \mathbf{0.581}$$

(c) The thermal efficiency is $\eta_{\text{th}} = \frac{\dot{W}_{\text{net}}}{\dot{Q}_{\text{in}}} = \frac{2287 \text{ kW}}{8555 \text{ kW}} = \mathbf{0.267}$

The second-law efficiency of the cycle is defined as the ratio of actual thermal efficiency to the maximum possible thermal efficiency (Carnot efficiency). The maximum temperature for the cycle can be taken to be the turbine inlet temperature. That is,

$$\eta_{\text{max}} = 1 - \frac{T_1}{T_3} = 1 - \frac{303 \text{ K}}{1144 \text{ K}} = 0.735$$

and $\eta_{\text{II}} = \frac{\eta_{\text{th}}}{\eta_{\text{max}}} = \frac{0.267}{0.735} = \mathbf{0.364}$

9-151 The exergy destruction associated with each of the processes of the Brayton cycle described in Prob. 9-73 is to be determined.

Analysis From Prob. 9-91, $q_{in} = 584.62 \text{ kJ/kg}$, $q_{out} = 478.92 \text{ kJ/kg}$, and

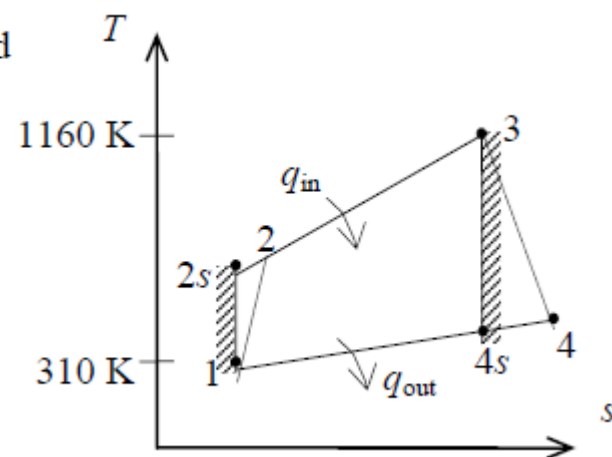
$$T_1 = 310 \text{ K} \longrightarrow s_1^\circ = 1.73498 \text{ kJ/kg} \cdot \text{K}$$

$$h_2 = 646.3 \text{ kJ/kg} \longrightarrow s_2^\circ = 2.47256 \text{ kJ/kg} \cdot \text{K}$$

$$T_3 = 1160 \text{ K} \longrightarrow s_3^\circ = 3.13916 \text{ kJ/kg} \cdot \text{K}$$

$$h_4 = 789.16 \text{ kJ/kg} \longrightarrow s_4^\circ = 2.67602 \text{ kJ/kg} \cdot \text{K}$$

Thus,



$$x_{\text{destroyed},12} = T_0 s_{\text{gen},12} = T_0 (s_2 - s_1) = T_0 \left(s_2^\circ - s_1^\circ - R \ln \frac{P_2}{P_1} \right) =$$

$$= (310 \text{ K}) (2.47256 - 1.73498 - (0.287 \text{ kJ/kg} \cdot \text{K}) \ln(8)) = \mathbf{43.6 \text{ kJ/kg}}$$

$$x_{\text{destroyed},23} = T_0 s_{\text{gen},23} = T_0 \left(s_3 - s_2 + \frac{q_{R,23}}{T_R} \right) = T_0 \left(s_3^\circ - s_2^\circ - R \ln \frac{P_3}{P_2} + \frac{-q_{in}}{T_H} \right)$$

$$= (310 \text{ K}) \left(3.13916 - 2.47256 - \frac{584.62 \text{ kJ/kg}}{1600 \text{ K}} \right) = \mathbf{93.4 \text{ kJ/kg}}$$

$$x_{\text{destroyed},34} = T_0 s_{\text{gen},34} = T_0 (s_4 - s_3) = T_0 \left(s_4^\circ - s_3^\circ - R \ln \frac{P_4}{P_3} \right) =$$

$$= (310 \text{ K}) (2.67602 - 3.13916 - (0.287 \text{ kJ/kg} \cdot \text{K}) \ln(1/8)) = \mathbf{41.4 \text{ kJ/kg}}$$

$$x_{\text{destroyed},41} = T_0 s_{\text{gen},41} = T_0 \left(s_1 - s_4 + \frac{q_{R,41}}{T_R} \right) = T_0 \left(s_1^\circ - s_4^\circ - R \ln \frac{P_1}{P_4} + \frac{q_{out}}{T_L} \right)$$

$$= (310 \text{ K}) \left(1.73498 - 2.67602 + \frac{478.92 \text{ kJ/kg}}{310 \text{ K}} \right) = \mathbf{220 \text{ kJ/kg}}$$

9-158 The exergy loss of each process for a regenerative Brayton cycle with three stages of reheating and intercooling described in Prob. 9-137 is to be determined.

Analysis From Prob. 9-137,

$$r_p = 4, \quad q_{\text{in},7-8} = q_{\text{in},9-10} = q_{\text{in},11-12} = 300 \text{ kJ/kg},$$

$$q_{\text{out},14-1} = 181.8 \text{ kJ/kg}, \quad q_{\text{out},2-3} = q_{\text{out},4-5} = 141.6 \text{ kJ/kg},$$

$$T_1 = T_3 = T_5 = 290 \text{ K}, \quad T_2 = T_4 = T_6 = 430.9 \text{ K}$$

$$T_7 = 556.7 \text{ K}, \quad T_8 = 855.2 \text{ K}, \quad T_9 = 575.5 \text{ K}$$

$$T_{10} = 874.0 \text{ K}, \quad T_{11} = 588.2 \text{ K}, \quad T_{12} = 886.7 \text{ K},$$

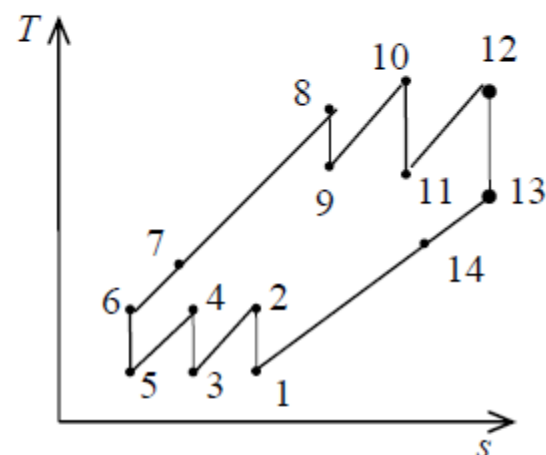
$$T_{13} = 596.7 \text{ K}, \quad T_{14} = 470.9 \text{ K}$$

The exergy destruction during a process of a stream from an inlet state to exit state is given by

$$x_{\text{dest}} = T_0 s_{\text{gen}} = T_0 \left(s_e - s_i - \frac{q_{\text{in}}}{T_{\text{source}}} + \frac{q_{\text{out}}}{T_{\text{sink}}} \right)$$

Application of this equation for each process of the cycle gives

$$\begin{aligned} x_{\text{dest},1-2} &= x_{\text{dest},3-4} = x_{\text{dest},5-6} = T_0 \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right) \\ &= (290) \left[(1.005) \ln \frac{430.9}{290} - (0.287) \ln(4) \right] = \mathbf{0.03 \text{ kJ/kg} \approx 0} \end{aligned}$$



$$x_{\text{dest}, 7-8} = T_0 \left(c_p \ln \frac{T_8}{T_7} - R \ln \frac{P_8}{P_7} - \frac{q_{\text{in},7-8}}{T_{\text{source}}} \right) = (290) \left[(1.005) \ln \frac{855.2}{556.7} - 0 - \frac{300}{886.7} \right] = \mathbf{27.0 \text{ kJ/kg}}$$

$$x_{\text{dest}, 9-10} = T_0 \left(c_p \ln \frac{T_{10}}{T_9} - R \ln \frac{P_8}{P_7} - \frac{q_{\text{in},9-10}}{T_{\text{source}}} \right) = (290) \left[(1.005) \ln \frac{874.0}{575.5} - 0 - \frac{300}{886.7} \right] = \mathbf{23.7 \text{ kJ/kg}}$$

$$x_{\text{dest}, 11-12} = T_0 \left(c_p \ln \frac{T_{12}}{T_{11}} - R \ln \frac{P_{12}}{P_{11}} - \frac{q_{\text{in},11-12}}{T_{\text{source}}} \right) = (290) \left[(1.005) \ln \frac{886.7}{588.2} - 0 - \frac{300}{886.7} \right] = \mathbf{21.5 \text{ kJ/kg}}$$

$$x_{\text{dest}, 8-9} = T_0 \left(c_p \ln \frac{T_9}{T_8} - R \ln \frac{P_9}{P_8} \right) = (290) \left[(1.005) \ln \frac{575.5}{855.2} - (0.287) \ln \left(\frac{1}{4} \right) \right] = \mathbf{-0.06 \text{ kJ/kg} \approx 0}$$

$$x_{\text{dest}, 10-11} = T_0 \left(c_p \ln \frac{T_{11}}{T_{10}} - R \ln \frac{P_{11}}{P_{10}} \right) = (290) \left[(1.005) \ln \frac{588.2}{874.0} - (0.287) \ln \left(\frac{1}{4} \right) \right] = \mathbf{0.42 \text{ kJ/kg} \approx 0}$$

$$x_{\text{dest}, 12-13} = T_0 \left(c_p \ln \frac{T_{13}}{T_{12}} - R \ln \frac{P_{13}}{P_{12}} \right) = (290) \left[(1.005) \ln \frac{596.7}{886.7} - (0.287) \ln \left(\frac{1}{4} \right) \right] = \mathbf{-0.05 \text{ kJ/kg} \approx 0}$$

$$x_{\text{dest}, 14-1} = T_0 \left(c_p \ln \frac{T_1}{T_{14}} - R \ln \frac{P_1}{P_{14}} + \frac{q_{\text{out},14-1}}{T_{\text{sink}}} \right) = (290) \left[(1.005) \ln \frac{290}{470.9} - 0 + \frac{181.8}{290} \right] = \mathbf{40.5 \text{ kJ/kg}}$$

$$x_{\text{dest}, 2-3} = x_{\text{dest}, 4-5} = T_0 \left(c_p \ln \frac{T_3}{T_2} - R \ln \frac{P_3}{P_2} + \frac{q_{\text{out},2-3}}{T_{\text{sink}}} \right) = (290) \left[(1.005) \ln \frac{290}{430.9} - 0 + \frac{141.6}{290} \right] = \mathbf{26.2 \text{ kJ/kg}}$$

$$\begin{aligned} x_{\text{dest,regen}} &= T_0 (\Delta s_{6-7} + \Delta s_{13-14}) = T_0 \left(c_p \ln \frac{T_7}{T_6} + c_p \ln \frac{T_{14}}{T_{13}} \right) \\ &= (290) \left[(1.005) \ln \frac{556.7}{430.9} + (1.005) \ln \frac{470.9}{596.7} \right] = \mathbf{5.65 \text{ kJ/kg}} \end{aligned}$$

Chapter 9

End of the Problems